MINI PROJECT 3: AI-ASSISTED CYBERSECURITY THREAT MODEL & TELEMETRY ANALYSIS

Course / Cohort: G17 CS - Module 1

Module: Cybersecurity Foundations, Networking & Ethical Hacking Lab

Student Name: Gaurav Singh Chouhan

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1. Executive Summary

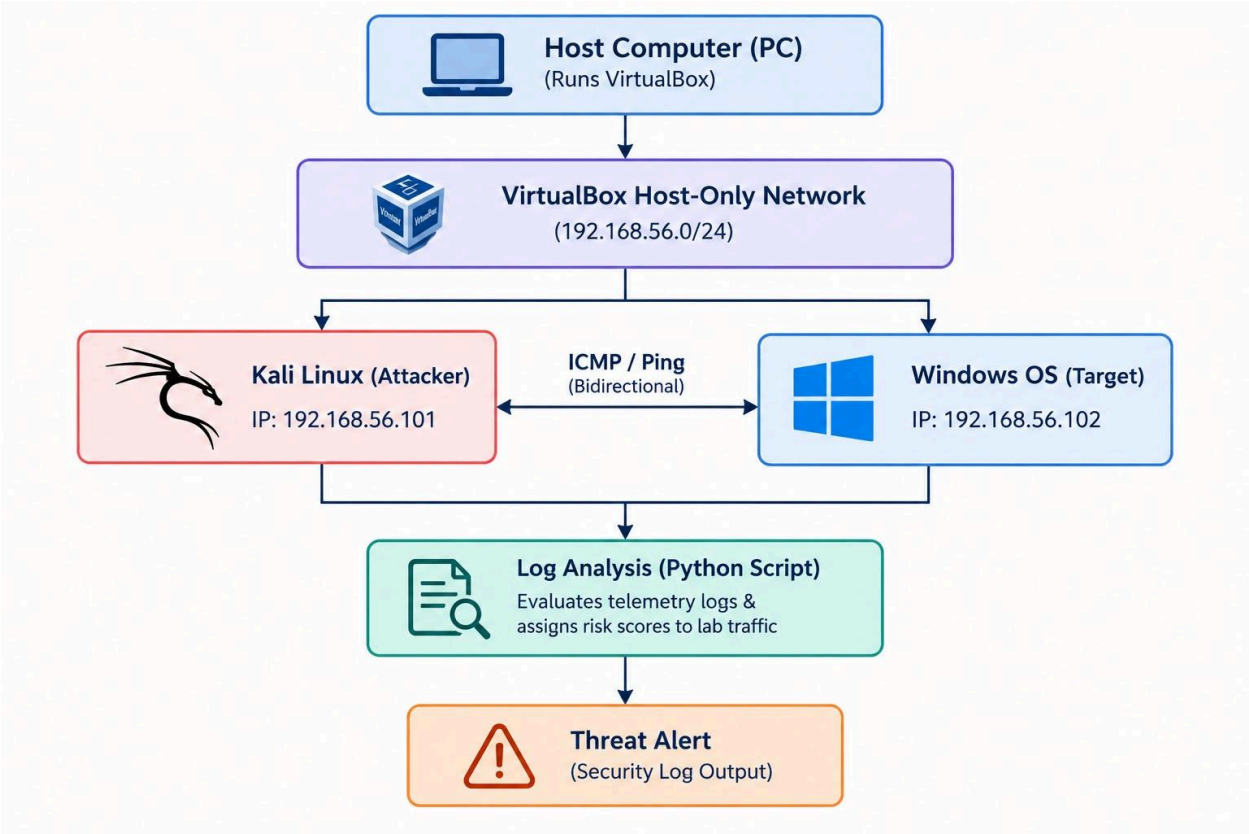
This project delivers a practical, AI-augmented threat modeling and log telemetry analysis for an isolated virtual laboratory environment (192.168.56.0/24). Tying directly into the dual-VM setup built during previous modules, this study establishes an architectural line diagram for the lab layout, details potential attack paths between the attacking and target nodes, and implements a lightweight Python-based anomaly detection engine to automatically evaluate and score network event logs.

2. Virtual Lab Scope & System Architecture

The analysis focuses on the Host-Only network segment connecting the attacker and target nodes within Oracle VM VirtualBox:

- **Attacker System (Kali Linux):** 192.168.56.101 (Interface: eth1)
- **Target System (Windows OS):** 192.168.56.102 (Interface: Host-Only Adapter)
- **Network Isolation:** 192.168.56.0/24 (Host-Only Subnet)

Architectural Line Diagram



3. STRIDE Threat Model Analysis

Threat vectors targeting the target Windows system (192.168.56.102) were evaluated using the STRIDE threat modeling framework:

- **Spoofing (S):** ARP poisoning/spoofing on the local subnet to intercept traffic between virtual adapters.

- **Tampering (T):** Payload modification of unencrypted layer-4 traffic in transit across the internal switch.

- **Repudiation (R):** Insufficient audit logging on the target OS masking port scanning or ICMP probe origins.

- **Information Disclosure (I):** Open default services (e.g., Port 135 RPC / Port 445 SMB) leaking host operating system details to network mappers like Nmap.

- **Denial of Service (D):** High-volume ICMP flooding or SYN-ACK queue exhaustion against host interfaces.

- **Elevation of Privilege (E):** Exploiting unpatched administrative services to escalate privileges from a remote low-privilege session.

4. Practical Implementation: Telemetry Analysis Script

To simulate automated threat detection within the lab, a custom Python threat analysis engine (lab_threat_model.py) was executed on the Kali Linux node. The script parses simulated network session logs, evaluates timing anomalies (off-hours execution), inspects action signatures (port scans and brute-force attempts), and calculates cumulative risk scores.

Script Execution & Output Verification



Figure 1: Execution output of lab_threat_model.py in Kali Linux terminal detecting anomalous lab events.

5. Risk Assessment & Engineering Controls

Threat Category	Detected Action	Risk Level	Recommended Engineering Control

Reconnaissance	Port 135 Scan	Mediu m	Enforce host firewall rules to restrict inbound RPC/SMB probes.
Authentication Abuse	Brute-Force Attempts	High	Implement local account lockout policies after 3 failed login attempts.
Network Probing	ICMP Echo Request	Low	Limit or drop ICMP ping responses on non-management interfaces.

6. Conclusion

Mini Project 3 successfully bridges theoretical threat modeling with practical execution. By pairing a STRIDE risk assessment with a custom telemetry analyzer script, this project demonstrates an end-to-end understanding of how security controls detect, score, and mitigate suspicious activity within virtualized network environments.